

Random Signal Theory–EEE 554

Lecture Notes 3

Two Jointly Distributed Random Variables

[Reading: Kay, Sections 7.3-7.5, 12.3-12.5]

Joint probability density function (PDF) and Properties:

Definition: $f_{XY}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{XY}(x, y)$, derivative of joint CDF

P1. $\int \int_{S_{X,Y}} f_{XY}(x, y) \, dx \, dy = 1 \Rightarrow$ volume under joint PDF is one

P2. PDF is non-negative, $f_{XY}(x, y) \geq 0$

P3. $\Pr(x_1 < X \leq x_2, y_1 < Y \leq y_2) = \int_{x_1}^{x_2} \int_{y_1}^{y_2} f_{XY}(x, y) \, dy \, dx$

P4. Marginal PDF: $f_X(x) = \int_{S_y} f_{XY}(x, y) \, dy$ and $f_Y(y) = \int_{S_x} f_{XY}(x, y) \, dx$

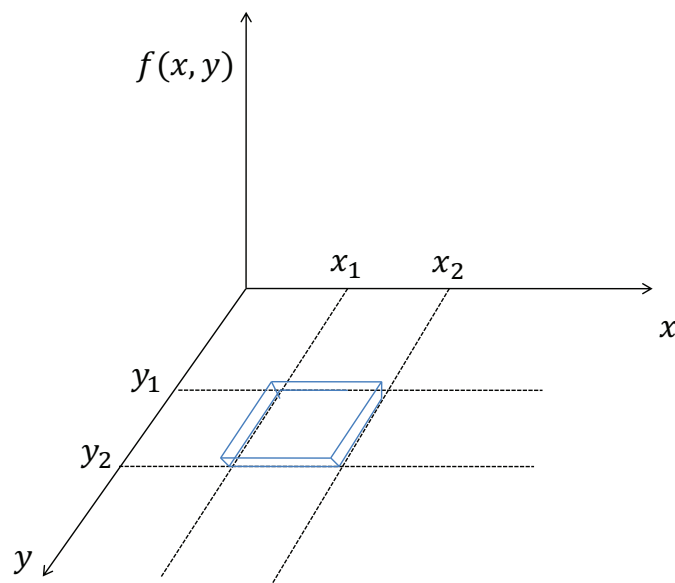
P5. Joint PDF is used to find probability of (x, y) points in a region:

$$\Pr((x, y) \in D) = \int \int_D f_{XY}(x, y) \, dx \, dy$$

Example:

Find marginal PDF of X given joint uniform PDF, $X, Y \sim \mathcal{U}(x \in (x_1, x_2), y \in (y_1, y_2))$

$$f_{XY}(x, y) = \begin{cases} \frac{1}{(x_2 - x_1)(y_2 - y_1)}, & x_1 < x < x_2, y_1 < y < y_2 \\ 0, & \text{otherwise} \end{cases}$$
$$= \frac{1}{(x_2 - x_1)(y_2 - y_1)} \left[\left(u(x - x_1) - u(x - x_2) \right) \left(u(y - y_1) - u(y - y_2) \right) \right]$$



To find the marginal

$$\begin{aligned} f_X(x) &= \frac{1}{(x_2 - x_1)} \int_{y_1}^{y_2} \frac{1}{(y_2 - y_1)} dy = \frac{1}{(x_2 - x_1)(y_2 - y_1)} y \Big|_{y_1}^{y_2} \\ &= \frac{1}{(x_2 - x_1)}, \quad x_1 < x < x_2 \end{aligned}$$

So, marginal of X is $X \sim \mathcal{U}(x_1, x_2)$, $f_X(x) = \frac{1}{(x_2 - x_1)} (u(x - x_1) - u(x - x_2))$

and similarly, $Y \sim \mathcal{U}(y_1, y_2)$

Classes of joint PDF:

Jointly distributed uniform random variables, $X, Y \sim \mathcal{U}(x \in (x_1, x_2), y \in (y_1, y_2))$

$$\text{Joint CDF } F_{XY}(x, y) = \begin{cases} 0, & x < x_1, \quad y < y_1 \\ \frac{(x - x_1)(y - y_1)}{(x_2 - x_1)(y_2 - y_1)}, & x_1 \leq x < x_2, \quad y_1 \leq y < y_2 \\ \frac{x - x_1}{x_2 - x_1}, & x_1 \leq x < x_2, \quad y \geq y_2 \\ \frac{y - y_1}{y_2 - y_1}, & y_1 \leq y < y_2, \quad x \geq x_2 \\ 1, & y \geq y_2, \quad x \geq x_2 \end{cases}$$

$$\text{Joint PDF } f_{XY}(x, y) = \begin{cases} \frac{1}{(x_2 - x_1)(y_2 - y_1)}, & x_1 < x < x_2, \quad y_1 < y < y_2 \\ 0, & \text{otherwise} \end{cases}$$

$$= \frac{1}{(x_2 - x_1)(y_2 - y_1)} \left[\left(u(x - x_1) - u(x - x_2) \right) \left(u(y - y_1) - u(y - y_2) \right) \right]$$

Example: Use PDF property P5 for $X, Y \sim \mathcal{U}(x \in (0, 1), y \in (0, 1))$ (see Example for E3) to find $\Pr(x_3 < X \leq x_4, y_3 < Y \leq y_4)$, where $x_3 = 0.5$, $x_4 = 2$, $y_3 = 0.25$, and $y_4 = 0.5$

Since $S_{X,Y} = \{x \in (0, 1), y \in (0, 1)\}$, the joint uniform PDF is

$$f_{XY}(x, y) = [(u(x) - u(x - 1))(u(y) - u(y - 1))]$$

Using the joint PDF and property P5,

$$\begin{aligned} \Pr(0.5 < X \leq 2, 0.25 < Y \leq 0.5) &= \int_{0.5}^2 \int_{0.25}^{0.5} f_{XY}(x, y) \, dy \, dx = \int_{0.5}^2 \int_{0.25}^{0.5} 1 \, dy \, dx \\ &= \int_{0.5}^2 \left(y \Big|_{0.25}^{0.5} \right) \, dx = (0.5 - 0.25) \left(x \Big|_{0.5}^2 \right) \\ &= (0.25)(0.5) = 0.125 \end{aligned}$$

Obtain the same value using CDF or PDF

$$\begin{aligned} \Pr(0.5 < X \leq 2, 0.25 < Y \leq 0.5) &= \int_{0.5}^2 \int_{0.25}^{0.5} f_{XY}(x, y) \, dy \, dx \\ &= F_{XY}(2, 0.5) + F_{XY}(0.5, 0.25) - F_{XY}(0.5, 0.5) - F_{XY}(2, 0.25) = 0.125 \end{aligned}$$

Jointly distributed Gaussian random variables, $X, Y \sim \mathcal{N}(\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)$

where $\rho = \frac{E[XY] - \mu_X \mu_Y}{\sigma_X \sigma_Y}$ is the correlation coefficient

► Joint Gaussian PDF

$$f_{XY}(x, y) = \frac{1}{2\pi\sqrt{(1-\rho^2)\sigma_X^2\sigma_Y^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left(\frac{(x-\mu_X)^2}{\sigma_X^2} - 2\rho\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2}\right)\right)$$

► If X and Y are jointly normal, marginals are also normal,

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2) \text{ and } Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

► If X is normal and Y is normal, it does not imply that X and Y are jointly normal

► If X and Y are IID (independent and identically distributed)

$$\mu_X = \mu_Y = \mu, \sigma_X^2 = \sigma_Y^2 = \sigma^2, \text{ and } E[XY] = E[X]E[Y] \Rightarrow \rho = 0, \text{ and PDF}$$

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{1}{2\sigma^2}\left((x-\mu)^2 + (y-\mu)^2\right)\right)$$

Independent RVs X and Y [Kay, Sections 7.5, 12.5]

If X and Y are independent, joint PDF (CDF) is product of marginal PDFs (CDFs)

$$F_{XY}(x, y) = F_X(x) F_Y(y), \quad f_{XY}(x, y) = f_X(x) f_Y(y)$$

Example: Joint exponential PDF of X and Y , $S_{X,Y} = \{x \in (0, \infty), y \in (0, \infty)\}$

(notation: $X, Y \sim \text{exp}$)

$$f_{XY}(x, y) = \begin{cases} e^{-x}e^{-y}, & x \geq 0, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

To determine whether X and Y are independent, find the marginal PDFs

$$\text{Marginal PDF of } X: f_X(x) = e^{-x} \int_0^{\infty} e^{-y} dy = e^{-x} \left(-e^{-y} \Big|_0^{\infty} \right) = e^{-x}, \quad x \geq 0$$

$$\text{Similarly for } Y, f_Y(y) = e^{-y} \int_0^{\infty} e^{-x} dx = e^{-y}, \quad y \geq 0$$

$\Rightarrow X$ and Y are independent since $f_{XY}(x, y) = f_X(x) f_Y(y)$

Other examples using joint exponential PDF, $X, Y \sim \text{exp}$,

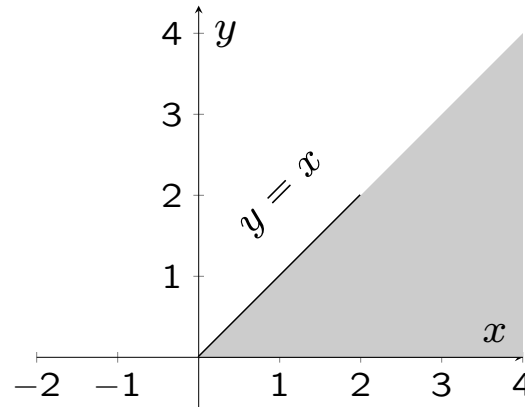
Find joint CDF given PDF **[see Kay, Example 12.3]**

Find joint PDF given CDF **[see Kay, Example 12.5]**

Use PDF, CDF to find probability of a region in (x, y) plane **[see Kay, Example 12.4]**

Example: Dependent RVs, joint sample space $S_{XY} = \{(x, y) : 0 \leq x < \infty, y \leq x\}$

$$f_{XY}(x, y) = \begin{cases} 2e^{-x}e^{-y}, & 0 \leq y \leq x < \infty \\ 0, & \text{otherwise} \end{cases}, \quad \text{sample space is shaded region}$$



Find the marginals to demonstrate that indeed X and Y are dependent

$$f_X(x) = \int_0^x f(x, y) dy = 2e^{-x} \int_0^x e^{-y} dy = 2e^{-x} \left(-e^{-y} \Big|_0^x \right) = 2e^{-x}(-e^{-x} + 1)$$

$$= \boxed{2e^{-x} - 2e^{-2x}, x \geq 0}$$

$$f_Y(y) = 2e^{-y} \int_y^\infty e^{-x} dx = 2e^{-y} \left(-e^{-x} \Big|_y^\infty \right) = \boxed{2e^{-2y}, y \geq 0}$$

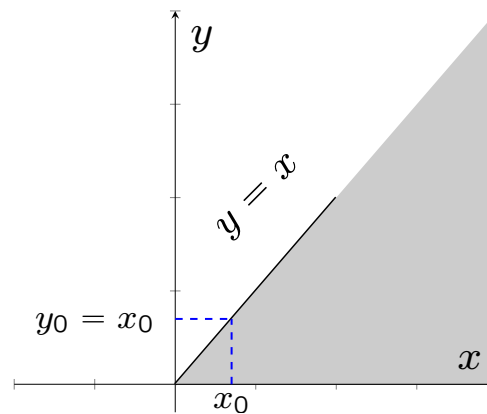
$$\Rightarrow f_{XY}(x, y) \neq f_X(x) f_Y(y) \Rightarrow \text{not independent}$$

Example:

Determine whether the given PDF is valid.

Sample space, $S_{XY} = \{(x, y) : 0 \leq x < \infty, y \leq x\}$ and PDF

$$f_{XY}(x, y) = \begin{cases} 2e^{-x}e^{-y}, & 0 \leq y \leq x < \infty \\ 0, & \text{otherwise} \end{cases}$$



If it is valid, then

$$\int \int_{S_{XY}} f_{XY}(x, y) \, dy \, dx = 1$$

Integration limits of outer integral (in terms of x) are $x = 0$ to ∞

Integration limits of inner integral (in terms of y):

$\Rightarrow$ fix $x = x_0$, find range of y in shaded region

$\Rightarrow$ limits of y are from 0 to y_0 , and $y_0 = x_0$ according to equation $y = x$

$$\begin{aligned}\int_{x=0}^{\infty} \left(\int_{y=0}^x f_{XY}(x, y) \, dy \right) dx &= \int_{x=0}^{\infty} 2e^{-x} \left(\int_{y=0}^x e^{-y} \, dy \right) dx \\&= \int_{x=0}^{\infty} 2e^{-x} \left(-e^{-y} \Big|_0^x \right) dx \\&= 2 \int_{x=0}^{\infty} e^{-x} (1 - e^{-x}) \, dx = 2 \int_{x=0}^{\infty} (e^{-x} - e^{-2x}) \, dx \\&= 2 \left(-e^{-x} \Big|_0^{\infty} \right) - \left(-\frac{1}{2}e^{-2x} \Big|_0^{\infty} \right) = 2(1 - 0.5) = 1\end{aligned}$$

$\Rightarrow$ valid joint PDF of X and Y